Studying and Enhancing Operational Efficiency at Mahak Spices Company

A Proposal report for the BDM capstone Project

Submitted by

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Declaration Statement

I am working on a Project titled "**Studying and Enhancing Operational Efficiency at Mahak Spices Company**". I extend my appreciation to **Mahak Spices Company**, for providing the necessary resources that enabled me to conduct my project.

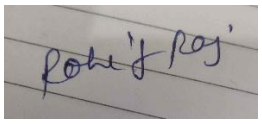
I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered from primary sources and carefully analyzed to assure its reliability.

Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the principles of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I understand that all recommendations made in this project report are within the context of the academic project taken up towards course fulfillment in the MSC Economics and Management Program offered by IIIT Lucknow.



Signature of Candidate

Name: Rohit Raj

Date: 22-08-2025

1 Executive Summary and Title (200 Words)

Mahak Spices Company is a small to medium scale spice trading shop located at Subhash Nagar, Shahjahanpur, Uttar Pradesh. The business is a B2B (Business to Business) and is engaged in wholesale trading of spices.

One of the challenges the business faces is its overdependence on a limited range of spices making it risky to fluctuations in demand or price, which can significantly impact overall performance. In addition to that, the possibility of spoilage during summer months poses a threat to product quality and inventory management. Lastly the fluctuating transportation costs continue to constrain on profit margins.

These challenges will be addressed through a comprehensive, data driven approach. This involves a detailed analysis of day-to-day sales and transportation expenses using data visualization and statistical tools to uncover trends and inefficiencies. By plotting the sales of each spice over time, the most and least popular spice can be identified, enabling to enhance the performance of underperforming items. Route optimization will be done to reduce the transportation costs. Additionally, to minimize spoilage, the price of perishable spices can be reduced to encourage quicker sales before degradation. By incorporating these strategies, the business will enhance profitability, ensure faster and more adaptable operations.

2 Organization Background (150 Words)

The business I am working with is Mahak Spices Company which is a wholesale spice trading business located in Subhash Nagar, Shahjahanpur, Uttar Pradesh. Established in August 2024 by Akshat and Mahak Jauhari, the business operates from a prime location shop that facilitates easy access to local trading market.

The shop primarily deals with B2B clients across the outskirts of Shahjahanpur, including regions such as Powayan, Jungbahadurganj, Pihani, Mohamdi and Shahbad. The timing of the shop is 9 AM to 7 PM and remains closed on Sundays. The shop consists of three dedicated staff members each responsible for marketing, delivery and shop management respectively.

As this is a B2B trading shop the customer base consists of business owners and traders aged between 30 and 67 years. Known for its high-quality products and strategic plannings the shop envisions becoming a leading and trusting wholesale spice supplier in the region.

3 Problem Statement (Listed as objectives) (50-75 Words)

3.1 Overdependence on few Spices

The business relies heavily on few spices leading to poor sales performance for other products. This leads to overstocking of unproductive items.

3.2 Fluctuating Transportation Costs

The involvement of third-party service providers causes significant variations in transportation costs. These unpredictable expenses contribute to degradation in the profit margins.

3.3 Spice Spoilage in Summer months

High temperature in summers leads to moisture loss in certain spices, resulting in reduced weight and compromised product quality. The spoilage affects both the profitability and customer satisfaction.

4 Background of the Problem (200 Words)

Overdependence on few Spices

The issue arises when the business focuses primarily on a few high demand spices while neglecting the promotion and distribution of other in the available stock. This not only limits the diversity of the product but also leads to underutilization of the inventory space and available stock. Therefore, results in spoilage of the products as well as depreciation of unsold product.

Fluctuating Transportation Costs

Variation in the transportation costs is mainly due to the variation in the delivery of the products each day. On certain day, the business may need to deliver large quantities of products, resulting in higher transportation expenses. These impact the overall profit for the day. The problem is further intensified by the involvement of third-party delivery services which vary their prices for delivery based on fuel prices, how far they have to travel and

whether the vehicle is available or not. Since there is no fixed delivery schedule and the amount to pay each day, it becomes difficult to manage profits properly.

Spice Spoilage in Summer months

During the summer months, high temperature and extreme heat can cause spices to lose moisture which results in the reduction of weight of the spices. So, if the spice sack was bought at a certain weight, it ends up weighing less by the time it's sold. This means that the business makes less money, as they are selling less than what they paid for. Also, this affects the quality of the spices as they may lose their freshness. This not only reduces the profits but also upsets the customer who expect fresh and good-quality spices.

5 Problem Solving Approach (400 Words)

Data Collection Process

To address and solve these problems, the very first step involves the process of collecting relevant data from the business for some amount of time (say 45 days). This data provides insights for analysis and helps identify the patterns and the area of improvement. The collected data gives the information about the following:

- **Daily Sales and Revenue –** The quantity of the product sold along with the cost per kg and the corresponding revenue.
- **Purchase cost –** The cost at which the product is purchased each day.
- **Transportation Cost –** The transport expenses per day.
- **Spice Type and Inventory –** The type of spice purchased and sold each day along with the quantities.

Once the data is collected, it is cleaned by replacing the empty cells with 0 and formatting the data to remove inconsistencies. Then preparing it for further analysis of profit trends, transportation costs trends, monthly price fluctuation trends of each spice.

Justification to collect the above data

By collecting daily sales and revenue data, the revenue trends over time can be plotted, as well as helps in calculating the total monthly revenue. It also helps in identifying the top selling spices including the percentage share of each spice in the overall revenue. Along with that the daily and monthly sales analysis can be done.

The purchase cost data is essential to calculate the profit trends over months.

Transportation cost data helps identifying the trend of the logistics-related expenses each day.

Spice Type and Inventory allows to address and analyze the issues of overstocking, and understanding the issue of spice spoilage and the demand patterns. It helps identifying the demand cycles of each spice and along with the popularity of that particular spice.

Analysis Tools

The analysis of the collected data will be done using

- MS-excel – for initial data entry, statistical calculations, pivot tables, visualization of the analyzed data by plotting charts.
- Python (Pandas) – for data cleaning and formatting.

6 Expected Timeline

6.1 Work Breakdown Structure:

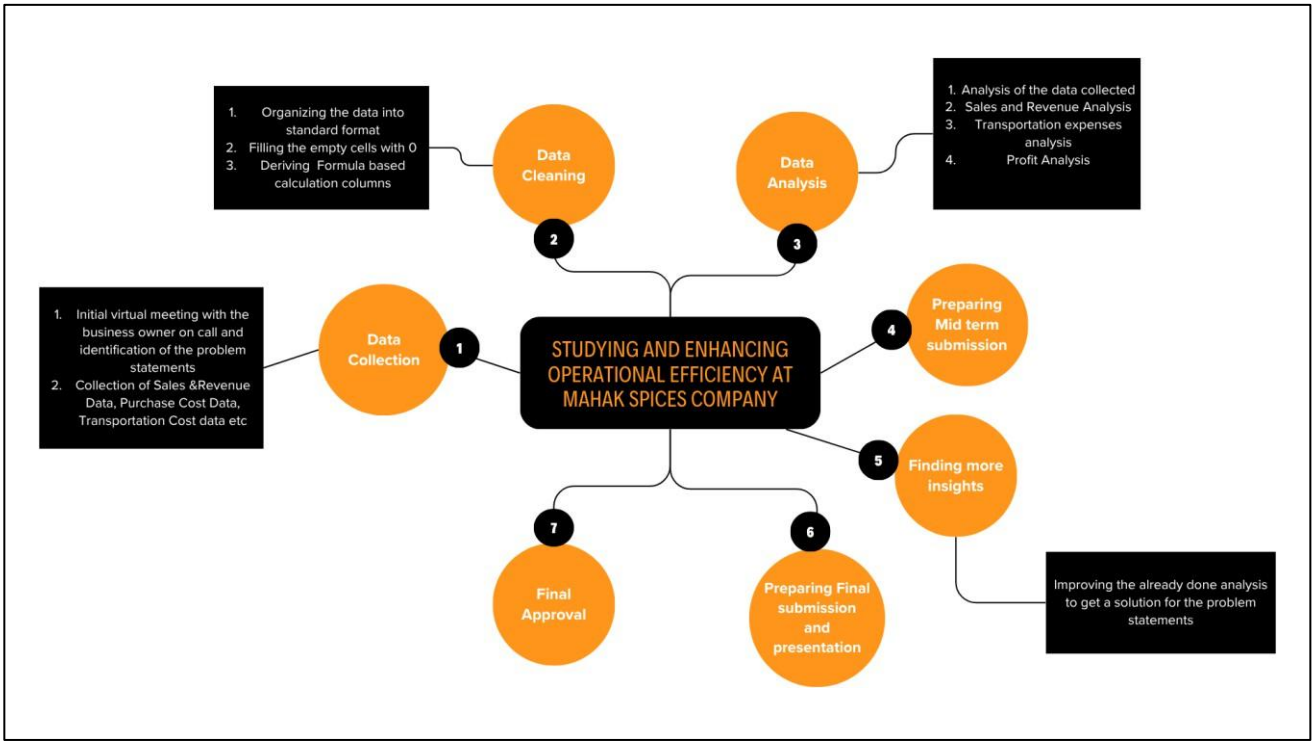


Figure 1- Work Breakdown Structure.

6.2 Gantt chart

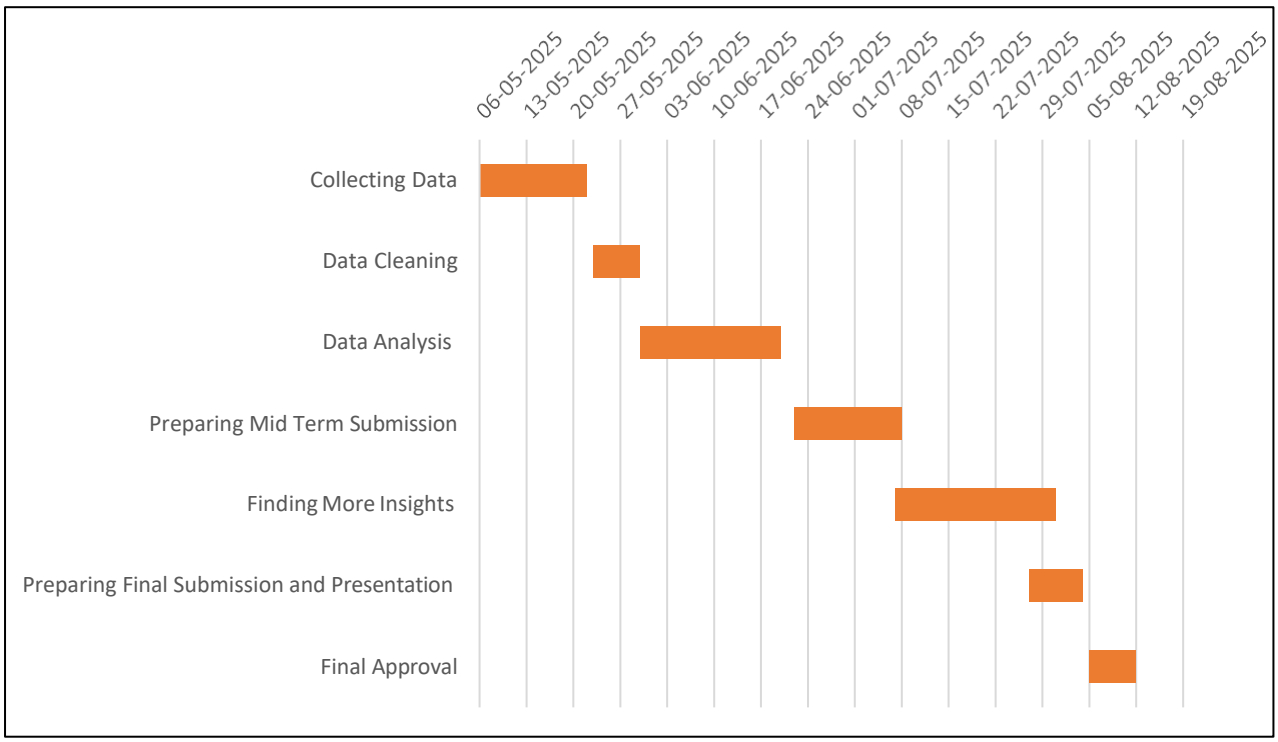


Figure 2 Gantt Chart

7 Expected Outcome (Briefly explain in 150-200 words)

7.1 Enhanced Net Profit

By analyzing daily sales, transportation costs, profit the business can spot the areas where the costs can be minimized and the profits can be maximized. This includes adjusting the cost of the spices and managing the plans more effectively. This can ultimately boost the profit of the business.

7.2 Recognition of Seasonal Sensitivity on Spice Quality

The spice inventory data helps identify the months which are risky to the spoilage of spices especially during summers. It allows the business to adopt better storage spaces and helps reduce the overstocking, maintain the overall product quality and minimize the loss.

7.3 Minimized Transportation Cost

By estimating the trends in transportation costs per day, the business can discover more cost-effective routes or shift to fixed cost partners which can reduce the daily fluctuations and timely deliveries of the product.